

Curriculum Vitae

Dr. Pankaj Chauhan (Ph.D.), JRF (BARC Funded Project)

Assistant Professor

Department of Chemistry

Madhav University, Pindwara

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Academic Qualification:

Degree	Subjects	Board/University	Name of Institute	Year
Ph.D.	Chemistry	GLA University (Mathura)	Inst. of Applied Sciences and Humanities	2024
M.Sc.	Chemistry	Jiwaji University, (Gwalior)	School of Study, Department of Chemistry	2019
B.Sc.	Chemistry, Zoology, Botany	University of Rajasthan (Jaipur)	Govt. P.G. College, Dholpur	2017
Intermediate	Physics, Chemistry, Biology, Hindi, English	Board of Secondary Education Rajasthan, Ajmer (Rajasthan)	Cheenu Public School Jakhi Dholpur (Rajasthan)	2014
High School	Hindi, English, Math, Social Science, Science, Sanskrit	Board of Secondary Education Rajasthan, Ajmer (Rajasthan)	Cheenu Public School Jakhi Dholpur (Rajasthan)	2012

Junior Research Fellow (JRF): Worked in the Department of Civil Engineering, GLA University Mathura as JRF in a project entitled “Probing Yamuna Water Pollution and Mechanism of its Diffusion to Groundwater using Isotope Hydrology” sanctioned by BRNS Mumbai, from September 28, 2023, to 9 August 2024.

List of Publications [Scopus/SCI/SCIE] (15):

1. Electrocatalytic properties of iron ferrite (Fe_3O_4) obtained by thermal decomposition method using egg-white (ovalbumin). *Russ. J. Electrochem.*, 2023, 59(4), 313–319. DOI: [10.1134/S1023193523040043](https://doi.org/10.1134/S1023193523040043)
2. Electrochemical oxygen evolution reaction on $\text{Ni}_x\text{Fe}_{3-x}\text{O}_4$ ($0 \leq x \leq 1.0$) in alkaline medium at 25°C. *J. Electrochem. Sci. Tech.*, 2022, 13(4), 497-503. DOI: <https://doi.org/10.33961/jecst.2022.00395>
3. Electrocatalytic oxygen evolution on GC/ $\text{Cu}_x\text{Fe}_{2-x}\text{O}_4$ ($0 \leq x \leq 1.0$) in 1M KOH. *Surf. Eng. Appl. Electrochem.*, 2023, 59(6), 754-763. DOI: [10.3103/S1068375523060054](https://doi.org/10.3103/S1068375523060054)
4. Electrocatalytic activity of Co_3O_4 for oxygen evolution in an alkaline medium at 25°C. *Indian J. Chem.*, 2023, 62, 1298-1302. DOI: [10.56042/ijc.v62i12.5901](https://doi.org/10.56042/ijc.v62i12.5901)
5. Electrochemical oxygen formation on $\text{Ni/La}_{1-x}\text{Sr}_x\text{CoO}_3$ ($0 \leq x \leq 1.0$) electrodes in alkaline medium at 25°C. *Russ. J. Electrochem.*, 2024, 60(8), 670-678. DOI: [10.1134/S1023193524700253](https://doi.org/10.1134/S1023193524700253)
6. Electrocatalytic activity of $\text{La}_{0.6}\text{Sr}_{0.4}\text{Cu}_y\text{Co}_{1-y}\text{O}_3$ for oxygen evolution in an alkaline medium at 25°C. *Indian J. Chem.*, 2024, 63, 875-881. DOI: [10.56042/ijc.v63i9.10838](https://doi.org/10.56042/ijc.v63i9.10838)
7. Electrocatalytic activity of Ni/LaCoO_3 for oxygen evolution in 1M KOH at 25°C obtained by alginic acid sol-gel method. *Indian J. Chem.*, 2025, 64(1), 93-97. DOI: [10.56042/ijc.v64i1.15027](https://doi.org/10.56042/ijc.v64i1.15027)
8. A comparative electrochemical performance of GC/ MFe_2O_4 ($\text{M} = \text{Cu} \& \text{Ni}$) electrodes for oxygen evolution in 1M KOH at 25°C. *Indian J. Chem.*, 2025, 64(4), 427-433. DOI: [10.56042/ijc.v64i4.17376](https://doi.org/10.56042/ijc.v64i4.17376)
9. Review on perovskite-type oxides and its applications in energy conversion and storage devices. *New J. Chem.*, 2025, 49(40), 17395-17419. DOI: [10.1039/d5nj01764e](https://doi.org/10.1039/d5nj01764e)

10. Electrocatalytic activity of GC/Ni_{0.5}Fe_{2.5}O₄ electrode in Alkaline Medium at 25°C. *Indian J. Chem.*, (**Accepted**)
11. Review on spinel type oxides towards oxygen evolution reaction. *Indian J. Chem. Technol.*, (**Accepted**)
12. Investigation of oxygen evolution reaction on perovskite-type oxides in alkaline media. *J. Sci. Ind. Res.*, (**Under Review**)
13. Microbial Innovations in Circular Wastewater Treatment for Enhanced Energy and Nutrient Recovery. *Indian J. Chem.*, (**Under Review**)
14. Breast Carcinoma and Hormone Receptors: Investigating Estrogen and Progesterone Expression for Enhanced Prognostic Insights. *Indian J. Chem.*, (**Under Review**)
15. Surface and Electrical Study of Fe-Fe-Substituted Bi-2223 Superconductor. *Indian J. Chem.*, (**Under Review**)

Book Chapters (2):

1. Adsorption Mechanisms of Hydrochar in Water and Wastewater Treatment. Springer Nature (**Under Review**)
2. Nanotechnology in Microbial Bioremediation. Taylor and Francis (**Under Review**)

Patents (1):

1. An apparatus for microbial fuel cells development with highly conductive electrodes and economical ion exchange. Application number 455508-001. Dr. Rohitash Kumar, Dr. Arun Kumar, Dr. Arun Kumar Kulshrestha, **Dr. Pankaj Chauhan**, Dr. Narendra Kumar Sharma, and Dr. Afsar Ali

International / National Conferences (16):

Workshops/Symposia Attended/orally presented:

1. 3rd International Conference on Nano-Architectures for Chemical, biological and Therapeutic Applications (NCBTA-2023) 24-26 November 2023 GLA University, Mathura.
2. 2nd International Conference on Nano-Architectures for Chemical, biological, and Therapeutic Applications (NCBTA-2022) 25-27 November 2022 GLA University Mathura.

3. International Virtual Conference on Modern Instrumentation and characterization techniques in Applied sciences-2020 (MICTAS-2020)- Uttarakhand science education and research center (U-SERC).
4. Worked as a committee member in the 4th International Conference on Futuristic and Sustainable Aspects in Engineering and Technology (FSAET-2023) 28-30 November 2023 at GLA University, Mathura.
5. National Intellectual Property Awareness Mission, Organized by Intellectual Property Office, India on 28 June 2023
6. Faculty development program on “Nanomaterials: Emerging Tools to Facilitate the World” Department of Chemistry GLA University Mathura 17 to 22 Oct 2022.
7. Faculty Development Programme on “Nanomaterials: Advance Applications” Department of Chemistry GLA University Mathura 27 September to 01 October 2021.
8. Faculty Development Programme on “Design and Fabrication of Miniaturized Biosensors” Organized by the Division of Healthcare Advancement, Innovation, and Research, Vellore Institute of Technology, Chennai, 09 to 11 October 2020.
9. National Level Faculty Development Programme on “Transforming Education for Sustainable Development: A Roadmap for Implementing NEP 2020 and SDGs in Higher Education Institutions, Jharkhand Rai University Ranchi, 21-27 January 2025.
10. International Faculty Development Programme on “Advanced Materials for Defence and Aerospace Applications” Organized by the Department of Mechanical Engineering, Ballari institute of Technology & Management Ballari Karnataka, 27 January to 1 February 2025.
11. Faculty Development Programme on “Manuscript Writing and Publication, Organized by the Department of Engineering, Madhav University Sirohi, Rajasthan, 23 June 2025.
12. Faculty Development Programme on “Outcome based Education(OBE) Strategies for Effective Implementation, Organized by the Department of Engineering, Madhav University Sirohi, Rajasthan, 16 July 2025.
13. National level Workshop on “Recent Trends in Agriculture and Science

with reference to Climate Change”, College of Agriculture, Madhav University Sirohi, Rajasthan, 13-15 October 2025.

14. National Intellectual Property Awareness Mission, Organized by Intellectual Property Office, India on 09 January 2025.
15. A two days hands-on online training workshop on organic and perovskite device modeling and simulation using SETFOS software from FLUXIM at 12 to 13 November 2025, IT Impulse Technology.
16. Two-Days Online Hands-on Workshop on “NanoDCAL–RESCU–SETFOS–LAOSS Semiconductor Device Simulation at 19-20 December 2025, IT Impulse Technology.

Research Area / Computational and Experimental Skills:

1. Electrocatalysis, fuel cell, green synthesis of nanoparticles for electrocatalytic applications.
2. Ms. Office (Ms. Word), spreadsheet (Ms. Excel), PowerPoint, Internet Web browsers.

Research Experience:

- ❖ 7 years (4 years during Ph.D. + 1 year JRF + 2 year at Madhav University, Sirohi, Rajasthan)

Expertise:

1. In synthesizing transition metal, mixed oxides with spinel and perovskite families are employed in the electrocatalysis of water, fuel cells, metal-air batteries and water analysis.
2. In interpreting data based on SEM, IR, XRD, TGA/DTA, CV, and Tafel.

Working on Software:

1. Origin 8.5
2. CHI (Electrochemical Work Station)
3. Bio Render
4. Artificial Intelligence (AI)

Teaching & Research Experience:

Organizations	Post Hold	Nature of Job	Duration	Leaving Reason
Madhav University Pindwara, Sirohi, (Rajasthan)	Assistant Professor	Teaching & Research	2024-26	Working
GLA University, Mathura, (Uttar Pradesh)	Junior Research Fellow (JRF)	Research	2023-24	Joined Madhav University

Personal Profile

- Date of Birth : 01-01-1998
- Gender : Male
- Nationality : Indian
- Marital Status : Married
- Religion : Hindu
- Language known : Hindi & English
- Hobbies : Reading books & playing Cricket

References:

- (1) Dr. Arun Kumar
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- (2) Dr. Basant Lal
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DECLARATION

I declare that all the above information given by me is true and correct.

(Dr. Pankaj Chauhan)